MAHARISHI VIDYA MANDIR GROUP OF SCHOOLS

ARTIFICIAL INTELLIGENCE-WORKSHEET

S.NO	CHAPTER	PAGE NO
1	AI REFLECTION PROJECT CYCLE AND ETHICS	2-6
2	DATA LITERACY	6-12
3	MATH FOR AI	12-16
4	GENERATIVE AI	16-17
5	PYTHON BASICS	22-28

MAHARISHI VIDYA MANDIR GROUP OF SCHOOLS

WORKSHEET-1 – AI REFLECTION PROJECT CYCLE AND ETHICS

I.ONE WORD ANSWERS:

1. Which of the following is not a domain of AI?

- Data Science b)Computer Vision
- c)Natural Language Processing d) Neural Network

2. The information extracted through data science can be used to make a decision

about it. (True/False)

3. _____ is a domain of AI-related to data systems and processes, in which the

system collects numerous data, maintains data sets and derives meaning/sense

out of them.

4. Which of the following is an example of data science?

- Social media websites b) Price Comparison websites
- Online Shopping websites d) Blog web sites

5. CV stands for _____.

6. The _____ is the domain of AI that depicts the capability of a machine to get

and analyse visual information and afterwards predict some decisions about it.

7. Which of the following translates digital visual data into descriptions and then

turned into a computer-readable language to aid the decision-making?

- Computer Vision b) PixelLab c) Pixel It d) Pixar

8. Which of the following is an example of a CV?

- Self-driving cars b)Smart Interactions c)Face Locks d)All of these

9. NLP stands for _____.

10._____ is a branch of artificial intelligence that deals with the interaction

between computers and humans using the natural language.

11.Which of the following is an example of NLP?

- Online translators b)Email filters c)Smart Assistants d) All of these

12.Which of the following component of NLP establish linkage with natural language

inputs and analyse different aspects of language?

- NLU b) NLG c) NLTK d) NLC

13._____ produces meaningful phrases and sentences in the form of natural

language.

14.The full form of NLU is _____

- Natural Language Understanding b)Natural Language Utilization
c) Natural Langauge Unit d)Natural Language Union

15.The full form of NLG is _____

a)Natural Language Group b)Natural Langauge Gap
c)Natural Language Generation d)Natural Language gaming

16. _____ can be defined as the science of moral duty and ideal human

behaviour that teaches what is right and what is wrong.

17. _____ refers to the basic principles of AI system design that use a good code of conduct and produce the results.

18. Which of the following can be a simple objective system that depends on human judgement?

- Logical Decision b)Logical Data c)Logical Input d)Logical Access

19. Who is accountable for considering the system impact on the world?

- The system b)Only the developer
- Only the company d)Every person involved in the creation

20. If a smart self-driving car met an accident with a person on the road, who is

responsible morally?

- The car owner b)The person c)The developer d)All of these

21. The major ethics concerns are

- issues, situations, choices b)decisions, speeches, actions
c)feelings, thoughts, expression d)All of these

22. AI ethics is a blanket term used for dealing with all the ethical issues related to

AI. (True/False)

23. Which of the more important concern related to data?

- How does AI work b)How does AI take decisions?
c)How AI collects data? d)How does AI represent data?

24. The world of AI revolves around _____.

25.Which of the following is a major source of data for major companies?

- Our vehicles b)Our Home c)Our lifestyle d)Our smartphone

26.If you do not allow the app certain permissions like storage, phone, camera or contacts, you normally cannot access the features of the app. (True/False)

27.Which of the following information will be collected when you open the newly

downloaded app on your mobile?

- Contact, Location, Camera, Storage b)Only contact and Camera

c)Only Location and Storage d)Any of the above depends on the app

28.The AI system only performs the tasks exactly what they made for. (True/False)

29.AI systems are biased because they take decisions from the data and training

material provided to them. (True/False)

30._____ refers to the guideline and system purposes should be very clear to

its users.

31.The system should be designed in such a manner that the data collection and

purpose should be unlimited.

32. The _____ stage begins with the identification of the problem in the AI project cycle.

33. Problem scoping helps in_____.

a)setting the goal b)identifying the problem
c)deeper understanding of the problem d)All of these

34. Which of the following W of 4Ws canvas problems helps in analyzing the affected people directly or indirectly?

a)Who b)what c)where d)why

35. The _____ are people who face the problem and would be benefited from the solution.

36. The w for _____ among 4Ws, determine the nature of the problem.

37. The w for what in 4Ws also gather the evidence to prove that the problem you have selected actually exists.

38. Which of the following refers to where among 4Ws canvas?

a)stakeholders b)nature of the problem

c)context/situation/location d)solution or benefits to the stakeholders

39. The _____ helps to summarize all the key points into one single template.

40. If there are errors in the problem scoping, it leads to failure of the project or delay in the project. (True/False)

41. In problem scoping, one should always consider the positive and negative aspects of the problem.

42. The _____ is second stage of AI project cycle.

43. Data acquisition refers to collect data through various activities. (True/False)

44. Which of the can be a piece of information or facts and statistics collected together for reference or analysis?

a)Data b)Problem statement c)Feedback d)All of these

45. To predict the output you must train the data for the AI project. (True/False)

46. In data acquisition the data which is provided as input is called _____.

47. We can provide any data for the prediction to get a better result. (True/False)

48. The type of data being collected in data acquisition is known as _____.

49. Which of the following is not a method to collect data?

a)Surveys b)Web Scrapping c)Sensors d)Archives

50. Which of the following is one of the government open source for collecting data?

a) www.ayushmanbharat.org b)www.india.gov.in

c)www.bharatdata.in d)www.india.com

51. The data which contain whole numbers categorized into which of the following categories?

a)Continuous numeric data b)non-continuous numeric data
c)Discrete numeric data d)uncountable numeric data

52. The data which can be in the form of any value within a range is considered as which of the following categories of data?

a)Continuous numeric data b)non-continuous numeric data
c)Discrete numeric data d)uncountable numeric data

53. Data has been organized according to some predefined rules, ideas or unorganized. (True/False)

54. The data which has predefined structure and organized in predefined manner is called _____ data.

55. The types of data that do not have any predefined structure are known as____ data.

56. The combination of structured and unstructured data is known as _____ data.

II.TWO MARK QUESTIONS:

1.What are the main approaches used for AI modelling?

2.Mention the ways to collect data.

3.Mention two websites from where we get open-source data.

4.What do you understand by data acquisition?

5.Explain the training data and testing data with an example.

6.Where do we collect data from?

7.What do you understand by data privacy?

8.What are various Model evaluation techniques?

9.What is deployment?

10.what is Evolution?

III.ANSWER IN DETAIL:

1.What can be done in each phase of the AI project cycle? Write points.

2. Create a 4W Project Canvas for the following.
As more and more new technologies get into play, risks will get more concentrated into a common network. Cybersecurity becomes extremely complicated in such scenarios and goes beyond the control of firewalls. It will not be able to detect unusual activity and patterns including the movement of data. Think about how AI algorithms can scrape through vast amounts of logs to identify susceptible user behavior. Use an AI project cycle to clearly identify the scope, how you will collect data, model and evaluation parameters.

MCQ ANSWERS

- Neural Network
- True
- DATA SCIENCE
- Price Comparison websites
- COMPUTER VISION
- CV
- Computer Vision
- All of these
- NATURAL LANGUAGE PROCESSING
- NLP
- . All of these
- NLU
- NLG
- Natural Language Understanding
- Natural Language Generation
- Ethics
- AI ETHICS

- Logical Decision
- Every person involved in the creation
- The developer
- decisions, speeches, actions
- TRUE
- How does AI take decisions?
- data
- Our smartphone
- True
- Any of the above depends on the app
- True
- True
- Transparent System
- False
- PROBLEM SCOPING
- All of these
- Who
- STAKE HOLDERS
- What
- True
- context/situation/location
- True
- True
- True
- Data Acquisition
- True
- Data
- True
- Training Data.
- False
- Data Features
- Archives
- www.india.gov.in
- Discrete numeric data
- Continuous numeric data
- True
- Structured
- un-structured
- semi-structured

MAHARISHI VIDYA MANDIR GROUP OF SCHOOLS

DATA LITERACY

I.MULTIPLE CHOICE QUESTIONS:

1. The most reliable source for news online is likely to be:

- A celebrity's social media post.
- A website with anonymous authors.
- A well-established news organization website.
- A blog with no editorial oversight.

2. Which of the following is NOT a benefit of data literacy?

- Making informed decisions about online purchases.
- Evaluating the credibility of news sources.
- Understanding complex scientific research.
- Being able to write computer code.

3. The final stage of the data literacy process framework involves:

a.Collecting data. b.Making data-driven decisions.

c. Analyzing data. d.Sharing findings in a complicated way.

4. When shopping online, data literacy can help you to:

- Choose the fastest shipping option regardless of cost.
- Compare prices and read reviews.
- Trust all online advertisements.
- Ignore product descriptions.

5. Two-factor authentication (2FA) adds an extra layer of security by requiring:

- Only using uppercase letters in your password.
- A secondary verification step in addition to your password.
- Changing your password every week.
- Downloading a free antivirus program.

6. Data that is organized into rows and columns is classified as:

a.Textual Data b.Structured Data

c.Numeric Data d.Unstructured Data

7. Which of the following is NOT a method for data pre-processing?

a.Error Correction b.Data Visualization

c.Formatting d.Remove Duplicates

8. What are the basic building blocks of qualitative data?

a.Individuals b.Units

c.Categories d. Measurements

9. Which among these is not a type of data interpretation?

a.Textual b.Tabular c.Graphical d.Raw data

10. A Bar Graph is an example of?

a.Textual b.Tabular c.Graphical d.None of the above

11. _____relates to the manipulation of data to produce meaningful insights. a.Data Processing b.Data Interpretation

c. Data Analysis d.Data Presentation

B. Fill in the Blanks

1. A strong password should be _____ and include a mix of characters.

2. Asking questions about the source and purpose of data is an example of

_____ thinking.

3. The data pyramid shows how raw data is transformed into _____ and ultimately _____.

4. The ability to access and retrieve data from various sources is called _____ literacy.

5. When data is organized and presented in a way that tells a story, it becomes

_____ .

6. The final stage of the data literacy process involves making decisions based on

the _____ of the data.

7.When you download software from a trusted source, you are practicing _____security.

8.Data that can be counted or measured is called_____data.

9.The process of gathering data is called_____.

10.Data that consists of words and phrases is_____data.

11.Data that can take any value within a range is_____data.

12.Ensuring people understand how their data is used is an aspect of data _____.

13.Removing entries with missing information is a data cleaning technique called _____.

14.Creating new data points from existing data is called_____.

15.Information about each piece of data within a dataset is called a _____.

c. Assertion and Reasoning.

1. Assertion (A): Data literacy is only useful for careers in data analysis. Reason (R): Data is used in many jobs, not just those focused on analysis.

use data literacy skills to determine if the information is likely to be true?

Identify A and R as True/False and explain why.

2. Assertion (A): All online reviews can be trusted as reliable sources of information. Reason (R): User reviews can provide valuable insights into products or services.

3. Assertion (A): Using a secure Wi-Fi network is the only way to protect your data online. Reason (R): There are several steps you can take to improve your online data security.

Identify A and R as True/False and explain why.

4. Assertion (A): Data security and data privacy are the same concepts. Reason (R): Both involve protecting information, but they focus on different aspects.

D.COMPETANCY BASED QUESTION:

1. You are planning a vacation and want to find a destination that aligns with your

interests. Describe how you would use data literacy skills to research different vacation spots.

2.You are tasked with collecting data on customer satisfaction for a new e- commerce website. What methods would you use and why?

3.Imagine you're building an AI model to predict flight delays. What data features would you consider including? Explain your choices.

4.What are some of the best practices for creating effective data visualizations?

5.What are the different types of data visualization tools available, and what are their strengths and weaknesses?

TWO MARK QUESTIONS:

1.What are some benefits of being data literate?

2.Describe the steps involved in the data literacy process framework

3.Explain how data literacy can be used to make informed choices about your diet and exercise habits.

4.Describe the difference between continuous and discrete data.

5.Explain two challenges of data acquisition for AI projects.

6.Why is it important to consider ethical principles when using data for AI development?

7.Briefly explain the concept of data augmentation.

8. How does data visualization differ from data analysis?

9.What are the advantages and disadvantages of using charts and graphs versus tables to present data?

10.

E14

	A	B	C	D	E	F
1	Song Name	Album	Artist	Genre	Year	Song Length
2	Blinding Lights	Blinding Lights	The Weekend	R&B	2019	03:20
3	Savage Love	Savage Love	Jason Derulo	Hip-Hop	2020	02:49
4	Watermelon Sugar	Fine Line	Harry Styles	Pop	2019	02:54
5	Happy	Happy	Pharrell Williams	Pop	2013	03:53
6	Panini	EP7	Lil Nas X	Hip-Hop	2019	01:55
7	Cake by the Ocean	DNCE	DNCE	Pop	2016	03:39
8	7 Rings	Thank you, next	Ariana Grande	Pop	2019	02:59
9	24K Magic	24K Magic	Bruno Mars	Funk	2016	03:46

- Is the Year qualitative or quantitative?
- Is Song Length discrete or continuous?

I the Genre discrete or continuous?

MCQ ANSWERS

1. (c) A well-established news organization website (Look for reputable sources with

editorial oversight)

2. (d) Being able to write computer code (Data literacy is about understanding and using data, not necessarily coding)

3. (c) Making data-driven decisions (This is the culmination of the data analysis

process)

4. (b) Compare prices and read reviews (Data literacy helps evaluate information for informed decisions)

5. (b) A secondary verification step in addition to your password. (2FA adds an extra layer of security beyond just a password

6. 1. (b) Structured Data

7. (b) Data Visualization (Data Visualization helps explore and understand data, not

pre-process it)

8. (c) Categories

9. (d) Raw data (Raw data is the unprocessed data before interpretation)

10. (c) Graphical

11. (c) Data Analysis

II.FILL UP

1.Long and include a mix of uppercase and lowercase letters, numbers, and symbols.

2. Critical thinking

3. Information, knowledge

4. Data access literacy

5. Information

6. Interpretation

7. Download

8. Numeric data

9. Data Acquistion

10. Textual

11. Continuous data

12. Transparency

13. Deletion

14. Data Augmentation

15. Data Feature

III. ASSERTION AND REASONING:

1. Assertion (False), Reason (True): Data literacy is beneficial in various fields, not

just data analysis. It helps interpret data in marketing, finance, healthcare, and even everyday life.

2. Assertion (False), Reason (True): Not all online reviews are reliable. Reviews can

be biased, fake, or incentivized. Data literacy skills help you evaluate the credibility

of reviews by checking for consistency, language patterns, and reviewer history.

3. Assertion (False), Reason (True): Secure Wi-Fi is important, but other steps include

strong passwords, 2FA, keeping software updated, and being cautious about

downloads and links.

4. Assertion (False), Reason (True): Data security focuses on protecting information

from unauthorized access, while data privacy is about controlling how your data is

collected, used, and shared.

IV. Competency Based Questions:

1. Planning a Vacation:

• Identify your interests (beaches, museums, adventure activities).

• Research destinations using travel websites and blogs (focusing on reputable sources).

• Compare data on weather, attractions, costs, and reviews.

• Analyze data to shortlist destinations that fit your interests and budget.

2. Data Privacy Presentation:

• Key Points: Explain what data privacy is, why it's important, and potential risks of

data breaches.

• Engagement: Use real-life examples, infographics, interactive polls, and case studies.

• Emphasize personal control: Explain how users can manage their privacy settings

on social media and other platforms.

3. By analyzing vast troves of data from sources like satellite imagery, radar, aircraft sensors, and weather stations, machine learning algorithms can identify patterns.

4. Best Practices for Effective Data Visualizations:

• Clarity and Simplicity: Ensure the chart is clear, concise, and easy to understand for the target audience.

• Accuracy: Represent data accurately and avoid misleading elements.

• Appropriate Chart Choice: Select the chart type best suited to convey the intended message (bar charts for comparisons, line charts for trends, etc.)

• Effective Use of Color and Labels: Choose color palettes that are visually appealing.

and accessible for colorblind viewers. Use clear labels and titles for all axes and

elements in the chart.

5. Data Visualization Tools:

• Tools (Strengths & Weaknesses):

o Tableau (Strength: Powerful for interactive dashboards, Weakness: Can be

complex for simple visualizations).

o Excel (Strength: Widely available, Weakness: Limited customization for

complex charts).

o Google Data Studio (Strength: Free and easy to use, Weakness: Fewer

customization options compared to paid tools).

o Power BI (Strength: Integration with Microsoft ecosystem, Weakness: Steeper

learning curve).

MAHARISHI VIDYA MANDIR GROUP OF SCHOOLS

MATH FOR AI

I.ONE WORD ANSWERS:

1.What is the main benefit of recognizing patterns in math and AI?

a.Improves memorization skills

b.Helps solve problems and make predictions

c.Makes learning more fun

d.Develops artistic abilities

2 .Which mathematical concept helps describe patterns using rules or formulas?

a.Geometry b. Statistics c.Probability d. Algebra

3. AI can learn patterns in what types of data?

a.Numbers only b. Images only c.Text only d. All of the above

4. AI excels at finding patterns in large amounts of data. What branch ofmathematics is most helpful in describing these patterns?

a.Algebra b. Calculus c.Geometry d. Statistics

5. In the series 2, 4, 6, 8, ..., what is the rule used to find the next number?

a.Add 1 b. Add 2 c.Multiply by 2 d. Divide by 2

6.Which of the following is NOT a common application of statistics in AI?

a.Identifying trends in customer behaviour b.Recognizing objects in images

c.Predicting the weather d.Solving systems of linear equations

7.The process of collecting, organizing, and analyzing data to draw conclusions iscalled?

a.Probability b.Statistics c.Calculus d.Algebra

8.Which mathematical concept helps calculate the likelihood of an eventhappening?

a.Geometry b.Statistics c.Probability d.Algebra

9.AI algorithms are like recipes that tell a computer what to do. Whatmathematical concept is similar to a recipe with steps?

a.Geometry b.Statistics c.Probability d.Algebra

10.In the image processing field of AI, what mathematical concept helps analyzethe arrangement of pixels in an image?

a.Geometry b.Statistics c.Probability d.Algebra

11.With one throw of a 6-sided die, what's the probability of getting an even number?

a.1/5 b.2/5 c.5/6 d. 1/2

12.Which of the following is an equation?

a.2x + 5 b. x + 2 = 4x c.x^2 + 2x d. 5 + 5x + 5x^2

13.What is the value of x? 10x-8=6x

a.8 b. 4 c.2 d. 6

14.What is the probability of getting head when you toss a coin once?

a.0.75 b.1 c.0 d. 0.5

15.The median of the data: 155, 160, 145, 149, 150, 147, 152, 144, 148 is

a.149 b. 150 c.147 d.144

II.FILL IN THE BLANKS

1.The ability to recognize arrangements of numbers, shapes, or colors is called_____.

2.AI uses_____solve complex problems by finding patterns in data.

3.Data used to train AI systems needs to be_____and_____- before itcan be used effectively.

4.A set of instructions that tell a computer what to do is called an_____.

5.When training an AI system, large amounts of_____are needed to help itlearn patterns.

III.MATCH THE FOLLOWING

	A	B
I.	Probability	a. exploring data
ii.	Calculus	b. finding out unknown or missing values
iii.	Statistics	c. predicting different events
iv.	Linear Algebra	d. training and improving AI model.

IV.TWO MARK QUESTIONS:

1.Explain two ways that math helps AI systems work.

2.Describe the role of statistics in the field of AI.

3.How does understanding patterns help us solve problems in math and AI?

4.Explain the relationship between Mathematics and Artificial Intelligence, providing justification for their interconnection.

5.Define Certain events and likely events with examples

6.Write any two examples of Impossible and equal probability events

V.LONG ANSWER QUESTIONS.

1.Identify the likely, unlikely, impossible and equal probability events from the following.

a.Tossing a coin

b.Rolling an 8 on a standard die

c.Throwing ten 5's in a row

d.Drawing a card of any suite

2.Explain the concept of "The Language of Numbers" in the context of AI.

3.Discuss the importance of math skills for someone interested in pursuing a career in AI development.

4.Choose a real-world application of AI (e.g., facial recognition software, self- driving cars) and explain how math is used in its development and operation.

VI.Scenario Based Questions.

1.Imagine you're training an AI to identify different types of flowers in pictures. How would you use math to achieve this?

2.A scientist is studying weather patterns to predict the likelihood of a hurricane. How might they use statistics and probability in their research?

VII.Competency Based Questions.

Age (in years)	10	12	14	15	16
Cases admitted (in a day)	5	7	9	22	11

1.Radhika collected the data of the age distribution of cases admitted during a day in a hospital. Find the average number of cases admitted in hospital. Also, draw a line graph to represent the data graphically.

2.You are tasked with designing a simple AI program that can identify basic shapes (circle, square, triangle) in images. How would you approach this problem using your understanding of math and patterns?

3.Imagine you're working on a team developing a new AI application for your school. What mathematical skills would be most valuable for you to contributeto the project? Explain your reasoning.

A. Multiple choice questions

1. B 2. D 3. D 4. D 5.B 6. D 7. B 8. C 9. D 10. A 11. D 12. B 13. C 14. D 15.

A

B. Fill in the Blanks:

1. Pattern 2. Math 3. Cleaned, Organized 4. Algorithm 5. Data

C. Match the following:

1. c) predicting different events

2. d) training and improving AI model

3. a) exploring data

4. b) finding out unknown or missing value

Competency based questions:

1.To find the average number of cases admitted, sum the number of cases and divide by the number of data points:

Average=5+7+9+22+115=545=10.8\text{Average} = \frac{5 + 7 + 9 + 22 + 11}{5} = \frac{54}{5} = 10.8Average=55+7+9+22+11=554=10.8

So, the average number of cases admitted in the hospital is **10.8**.

2.

1. Understanding the Mathematical Properties of Shapes
2. Data Collection
3. Feature Extraction
4. Model Training
5. Model Testing and Validation
6. Fine-tuning and Deployment

3.

1. Linear Algebra
2. Probability and Statistics
3. Calculus
4. Discrete Mathematics

5. Geometry

MAHARISHI VIDYA MANDIR GROUP OF SCHOOLS

GENERATIVE AI

CLASS:IX

A.Multiple Choice Questions (MCQs)

1.Generative AI can create new data in the form of

- Text only b. Images only
- Text, Images, Code and Music d. Numbers only

2. Generative AI models are trained on :

a.Existing User Data b.Instructions
c.Existing Data d.Random patterns

3.Generative Adversarial Networks (GANs) are a type of :

a. Search Engine b.Generative AI Model
c. Data Compression Tool d.Social Media Platform

4.Which of the following is not a benefit of Generative AI?

a. Increased Efficiency b.Boosted Creativity
c. Data Bias d.Personalized Experiences

5.Generative AI is limited by:

- User Interface Complexity b.High Availability
- Computational Demands d.Limited Internet Access

6.Popular Generative AI Tools include:

a.Artbreeder b.Runway ML

c.ChatGPT d.All of the above

7.Generative AI can be used to:

a.Improve medical Diagnosis b.Write fake news articles
c. Neither a nor b d.Both a and b

8.Generative AI models work by:

a.Following a set of pre-programmed rules

b.Identifying patterns in data

c.Randomly generating content

d.Connecting to the internet

II.Fill in the blanks

1. Generative AI is a type of artificial intelligence that can create new _____
2. The process of generative AI model involves feeding it a large amount of _____
3. Generative AI models can be used to _____ creativity.

4. One of the limitations of Generative AI is _____ bias in the training data.
5. Popular generative AI tools include _____ for creating artistic images.
6. Generative AI has the potential to _____ many industries.
7. A generative AI model _____ can generate realistic images.
8. Generative AI is a powerful tool, but it is important to use it _____.

A. Multiple Choice Questions (MCQs)

1. C 2. C 3. A 4. C 5. C 6. C 7. A 8. B

B. Fill in the Blanks

1. Data 2. Data 3. Boost 4. Data 5. Artbreeder 6. Revolutionize 7. Generative Adversarial Network (GAN) 8. Resposibly

MAHARISHI VIDYA MANDIR GROUP OF SCHOOLS

PYTHON BASICS

I.ONE WORD ANSWERS:

1. Which keyword is used for function in Python language?

a) Function

b) def

c) Fun

d) Define

2. Which of the following character is used to give single-line comments in Python?

a) // b) # c) ! d) /*

3. What will be the output of the following Python code?

i = 1

while True:

```
if i%3 == 0:
    break
print(i)
 i + = 1
```

a) 1 2 3 b) error c) 1 2 D)) none of the mentioned

4. Which is the correct operator for power(x^y)?
a) X^y b) X**y c) X^^y d) None of the mentioned

5. Which one of these is floor division?
a) / b) // c) % d) None of the mentioned

6. What is the answer to this expression, 22 % 3 is?
a) 7 b) 1 c) 0 d) 5

7. Mathematical operations can be performed on a string.
a) True
b) False

8. Which of the following is an invalid statement?
a) abc = 1,000,000
b) a b c = 1000 2000 3000
c) a,b,c = 1000, 2000, 3000
d) a_b_c = 1,000,000

9. Which of the following cannot be a variable?
a) __init__ b) in c) it d) on

10. Find the invalid variable among the following:

a. 1st_string b. my_string_1 c. _ d. foo

11. The order of precedence in the Python language is:

A) Exponential

B) Parentheses

C) Division

D) Multiplication

E) Subtraction

F) Addition

a. B,A,D,C,F,E

b. A,B,D,C,F,E

c. A,B,C,D,E,F

d. B,A,D,C,E,F

12. Which one of these is incorrect?

a. float('nan') b. float('inf') c. float('12+34') d. float('56'+'78')

13. The value of the Python expression given below would be: 4+2**5//10

a. 77 b. 0 c. 3 d. 7

14. Which arithmetic operators can we NOT use with strings?

a. – b. + c. * d. All of the above

15. Who developed Python Programming Language?
a) Wick van Rossum
b) Rasmus Lerdorf
c) Guido van Rossum
d) Niene Stom
16.. Is Python case sensitive when dealing with identifiers?
a) no b) yes c) machine dependent d) none of the mentioned

17. Which of the following is the correct extension of the Python file?
a) .python b) .pl c) .py d) .p

18. Is Python code compiled or interpreted?
a) Python code is both compiled and interpreted
b) Python code is neither compiled nor interpreted
c) Python code is only compiled
d) Python code is only interpreted

19. All keywords in Python are in _____
a) Capitalized b) lower case c) UPPER CASE d) None of the mentioned

20. What will be the value of the following Python expression? 4 + 3 % 5
a) 7 b) 2 c) 4 d) 1

21. Which of the following is used to define a block of code in Python language?
a) Indentation b) Key c) Brackets d) All of the mentioned
22. Which of the following character is used to give multie-line comments in Python?

a) // b) # c) , d) single quotes ''' or double quotes "

23.What is the extension of the Python file?
1) .pl 2).py 3).python 4).p
24.All the keywords in Python are in_

a.Lower case b.Upper case c.Capitalized d. None of the above
25.Write the output for the following program

```
principle_amount = 2000
roi = 4.5
time = 10
simple_interest = (principle_amount * roi * time)/100
print("datatype of principle amount : ", type(principle_amount))
print("datatype of rate of interest : ", type(roi))
print("value of simple interest : ", simple_interest)
print("datatype of simple interest : ", type(simple_interest))
```

26.write the output for the following program

```
c = 'Ram'
N = 3
print(c*N)
```

27.write the output for the following

```
a = 20
b = "Apples"
print(str(a)+ b)
```

28.What will be the value of the following Python expression? 4 + 3 % 5
a) 7 b) 2 c) 4 d) 1
29. Which type of Programming does Python support?
a) object-oriented programming b) structured programming
c) functional programming d) all of the mentioned

II.TWO MARK QUESTIONS

Q1) What are the key features of python ? Mention various applications along with it

Q2) What are the different modes for coding in python?

Q3) What are comments in python ? List down the various types of comments.

Q5) What are the rules for naming of variables and constants

Q6) Explain python input and output with the help of an example.

MCQ ANSWERS:

1.def
2. #
3. error
4. X**y
5. //
6. 1

7. False
8. a b c = 1000 2000 3000
9. *in*
10. *1st_string*
11. B,A,D,C,F,E
12. float('12+34')
13.7
14. –(minus)
15. Guido van Rossum
16.Yes
17. .py
18. Python code is both compiled and interpreted
19. lower case(except None,True,False)
20. 7
21. Indentation
22. single quotes ''' or double quotes "
23).py
24. Lower case
25. value of simple interest : 900
26.RamRamRam
27.20 Apples
28.7
29. all of the mentioned

MAHARISHI VIDYA MANDIR GROUP OF SCHOOLS

EMPLOYABILITY SKILL

1. What do you mean by close-ended questions?
a. Questions that can have any answer b. Questions that do not have answers
c. Questions with yes/no answers d. Questions that have many answers
2. Some answers are not limited or closed, these questions are called _____.
a. Open ended b. Close ended
c. Short ended d. None of the above
3. Which question should you ask if you don't understand a task you've been given?
a. Where are the reports of this task?
b. Is it possible for you to repeat the instructions for this task?
c. Is it possible for you to give example for this task?
d. Why are you doing this task?
4. _____ is she not talking to anyone in the class?
a. How b. Where c. Why d. What

5. _____ does she live?
a. Who b. Where c. When d. What
6. We can learn more by asking questions
a. Gain new knowledge, get information b. Make sure that what we know is correct
c. Avoid doubts, confusion, misunderstanding d. All of the above
7. When _____, you say 'Good Afternoon'.
a. 10 am b. 11.59 am c. 6 pm d. 1 pm
8. There are two methods of greeting _____ and _____.
a. Formal greetings and Informal greetings
b. Direct greetings and Indirect greetings
c. Both a) and b)
d. None of the above
9. The study of the sounds we make when we speak is known as _____.
a. Pronunciation b. Phonetics c. Sound d. Words
10. n the line "The children played football." identify the subject.
a. The children b. Children played c. Played d. Football
11. Sentences where the subject performs an action are called as _____.
a. Active voice b. Passive Voice c. Indirect Voice d. None of the above
12. What is a sentence?
a. It is collection of group of ideas
b. A collection of words that expresses a whole thought
c. A set of guidelines that must be followed in order to write properly
d. All of the basic punctuation signs and contained in this set of words.
13. A _____ is a word substituted in the place of a noun.
a. Noun b. Pronoun c. Adjective d. Verb
14. In the English language, there are _____ basic parts of speech.
a. 6 b. 7 c. 8 d. 9
15. Choose the most acceptable oral communication example.
a. Face to Face conversation b. Talking on a phone
c. Classroom teaching d. All of the above
16. Self management is also useful in:
a. Reaching your goals b. Overcoming bad habits
c. Creating positive habits d. All of the above
17. Which one is not applicable for self-management?
a. Understanding self b. Bargaining
c. Problem solving d. Confidence building
18. If you are aware of your own values, likes, dislikes, strengths, and shortcomings as an individual. It denotes the fact that you are _____.
a. Self-Confident b. Self-Control c. Self Motivated d. Self-Aware

19. Which one is not a good example of strength?
a. I am good at understanding other peoples b. Dealing with strangers, I am confident.
c. I don't know, how to play chess. d . I help my parents in household chores.
20. You can develop your self – confidence through changes in _____.
a. Attitude b. Practice c. Both a) and b) d. None of the above
21. Dipa Karmakar having flat foot since she was six years old, she work hard and become the first Indian female gymnast in the Olympics.
a. Self – belief b. Hard Work c. Positive Attitude d. Commitment
22. Mahatma Gandhi, the leader of the Indian Freedom Struggle, was a staunch supporter of India's independence from the British Raj. To gain India's freedom, he used nonviolent civil disobedience.
a. Self – belief b. Hard Work c. Positive Attitude d. Commitment
23. When the team wins a competition, congratulate them on their accomplishments. When the team loses a tournament, articulate learning.
a. Appreciate achievements & accept failures b.Have a goal and take steps towards it
c. Always look at the good side and be happy d. None of the above

24. What steps should one take to boost their confidence?
a. Set goals in life b. Appreciate oneself for all the achievements
c. Always think positively d. Talk to people who are confident
25. Which of the following is a positive thinking example?
a. Overcoming challenges
b. Making you do well or making you an energetic individual
c. Helping you get better at work
d. All of the above
26. _____ is beneficial to our health and well-being.
a. Cleanliness b. Regular School c. Work at home d. None of these
27. Personal Grooming is a term used to describe the act of _____.
a. Maintaining personal hygiene for pleasing personality b. Brushing teeth
c. Trimming nails d. None of these
28. t is very important to dressing and grooming properly, because they help us to look_____.
a. smart b. untidy c. shabby d. All of the above

29. Dressing and grooming are vital since they assist us in our daily activities.
a. Look smart b. Feel confident about ourselves
c. Make a good impression of ourselves d. All of the above
30. Radha uses a lot of hair oil in order to grow her hair. She does not wash her hair for days at a time, and it often stinks.. What would you advice her to do?
a. She can keep thee oil in her hair since it promotes hair growth.
b. She can wear it at night and wash her hair before leaving the house every day.
c. She shouldn't use any of the oil
d. To lesser the odour, she might apply the oil and a little water to her hair before leaving the house.
31. ICT stands for _____.
a. Information and Communication Technology
b. Informed and Communicated Technology
c. Informal and Communication Technology
d. None of the above
32. The ICT skills that you need are
a. Knowing how to operate computer
b. Knowing how to browse the Internet for collecting
c. Storing and dissemination information
d. All of the above
33. Imagine your friend sends you an e-mail. The e-mail will appear in your _____.
a. Inbox b. Trash c. Draft d. None of the above
34. The _____ can or folder holds e-mails that you no longer need and have deleted.
a. Trash b. Draft c. Junk/Spam d. None of the above
35. The _____ folder holds e-mails that may not be useful.
a. Trash b. Drafts c. Junk/Spam d. None of the above
36. If you want to receive mail from other person, you need to _____ the message.
a. Forward b. Reply To c. Reply All d. Copy
37. You can use the _____option to save an incomplete email without sending it.
a. Save as Draft b. Sent Items c. Trash d. Inbox
38. Which of the following is a popular E-Mail provider.
a. Gmail b. Yahoo c. Outlook d. All of the above
39. A message delivered through the Internet from one person to another is known as electronic mail. First part is known _____ and the second part is known as _____.

a. User name & Domain name b. Name & address
c. Name type & address d. None of the above
40. You are not seeing the entire page correctly. By pressing the button, you can reload the page.
a. Back Button b. Refresh Button c. Reset Button d. None of the above
41. It is a vast network of linked hypertext files, stored on computers throughout the world, that can provide a computer user with information on a huge variety of subjects.
a. World Wide Web b. Browser c. Search Engine d. None of the above
42. To connect to the Internet, the computer has to be connected to the _____.
a. Internet Society b. Internet Architecture
c. Internet Service Provider d. Large Area Network
43. 1 Kbps (Kilobits per second) = _____.
a. 1000 bits per second b. 2000 bits per second
c. 1024 bits per second d. 3000 bits per second
44. In the analog devices the bandwidth is expressed in _____.
a. Cycles Per Second b. Hertz (Hz) c. Both a) and b) d. None of the above
45. _____ helps to store all the information in the computer system.
a. Files b. Folder c. Image d. None of the above

46. People live together in villages, cities, states and countries, thus forming a _____.
a. Group b. Society c. Both a) and b) d. None of the above
47. Water covers about _____ of Earth's surface and is a necessary element for life.
a. One Quarter b. Two Quarters c. Three Quarters d. Four Quarters
48. There are _____ fundamental sources of energy
a. 2 b. 3 c. 4 d. 5
49. The resources which cannot be exhausted by human consumption are called _____.
a. Inexhaustible Resources b. Exhaustible Resources
c. Renewable Resources d. None of the above resources
50. _____ systems use resources that are constantly replaced and are usually less polluting.
a. Inexhaustible Resources b. Exhaustible Resources
c. Renewable Resources d. None of the above resources
51. _____ are the Non-renewable resources which cannot easily be replaced once they are destroyed.

a. Fossil fuels b. Minerals c. Both a) and b) d. None of the above

52. Only about _____ of water on earth is freshwater.

a. 5 % b. 2.5% c. 3% d. 3.5%

53. It is created by the decomposition of buried dead animals and plants in lakes and oceans. It's found in oil wells above the oil. Vehicles are powered by compressed _____ .

a. Natural Gas (CNG) b. Biomass

c. Nuclear energy d. All of the above

54. Which activities are damaging our earth and environment

a. Overexploitation b. Mining

c. Deforestation & Pollution d. All of the above

55. Pollutants that break down into simpler, harmless substances naturally (by the action of moisture and bacteria) are called _____.

a. Non-Biodegradable Pollutants b. Biodegradable Pollutants

c. Natural Pollutants d. None of the above

56. Because of the use of fossil fuels, the earth is becoming hotter (coal, petrol, diesel, etc.). Global warming is caused by these gases, which trap and prevent the earth's heat from leaving. This is known as _____.

a. Greenhouse Effect b. Red house Effect

c. Brown house Effect d. None of the above

57. **_____ are the main mechanism for the conversion of carbon dioxide into carbon and oxygen.**

a. Human b. Forest c. Water d. None of the above

58. Carbon dioxide in the atmosphere has increased by _____ since pre-industrial times, causing more heat to be trapped in the lower atmosphere.

a. 45 % b. 31 % c. 10 % d. 05 %

59. Cleaning chemicals, coolants in refrigerators and air conditioners, etc., release ozone depleting substances, such as _____ in the atmosphere.

a. Chlorofluorocarbons b. Carbons

c. Carbon dioxide d. None of the above

60. What are some of the environmental effect caused by the modern agricultural methods?

a. Chemical pollution due to fertilizers b. Improvement in the environment

c. Lower air pollution due to crops d. Decrease in forest areas

61. An _____ is a person who is self – employed, is willing to take a calculated risk and brings in a new idea to start a business.

a. Software Engineer b. Entrepreneurship

c. Civil Engineer d. Mechanical Engineer

62. What is the aim of entrepreneurship?

a. Solve customer' need b. Earn a profit

c. Both a) and b) d. None of the above

63. Entrepreneurs create _____ opportunities, to grow the economy of the country.
a. Business b. Credit c. Money d. None of the above
64. A business started by an entrepreneur is usually a very_____ in the beginning.
a. Large Scale of Business b. Small business
c. Modern businesses d. None of the above
65. It is important for an entrepreneur to be _____ in running a business.
a. Patient b. New Idea c. Both a) and b) d. None of the above
66. When a person begins a business, he or she may encounter several challenges and failures. Even while taking a significant risk, an entrepreneur must think _____.
a. Positively b. Negatively
c. Both a) and b) d. None of the above

67. The ability to handle take risks is a _____
a. Quality of Entrepreneur b. Role of Entrepreneur
c. Characteristics of Entrepreneurship d. None of the above
68. Entrepreneurs need to be open to _____and try them without fear of failure.
a. Invest all of money b. New Ideas
c. Both a) and b) d. None of the above
69. Experiment with many ideas is a quality related to _____.
a. Perseverance b. Confidence c. Creativity d. Trial and error
70. Entrepreneur's creativity and innovation quality are related to _____
a. Positive b. Patience c. Solution d. Hard Working
71. The phrase "green economy" was first coined in a _____ report for the UK government, titled Blueprint for a Green Economy, by a group of top environmental economists.
a. 1980 b. 1989 c. 1990 d. 1995
72. The concept of green growth aims at achieving economic growth that is socially inclusive and environmentally sustainable.
a. Environmental Growth b. Economic Growth
c. Country Growth d. None of the above
73. United Nations Sustainable Development Summit launched in New York in the year of _____.
a. 2010 b. 2012 c. 2015 d. 2020
74. Which of the following in NOT a goal for sustainable development that will be met by 2030?
a. Gender Equality b. Good Health and Wellbeing
c. Space Research d. Zero Hunger

75. Which of the following are non-renewable resources?(Choose. All options that apply)

a. Coal b. Diesel c. Sun d. Water

MULTIPLE CHOICE QUESTIONS ANSWERS:

1 . Questions that have many answers

2. Open ended

3. Is it possible for you to repeat the instructions for this task?

4.Why

5. Where

6. All of the above

7. 1 pm

8. Formal greetings and Informal greetings

9. Phonetics

10. The children

11. Active voice

12. A collection of words that expresses a whole thought

13. Pronoun

14. 8

15. Face to Face conversation

16. All of the above

17. Bargaining

18. Self-Aware

19. I don’t know, how to play chess

20. Both a) and b)

21. Hard Work

22. Commitment

23.Appreciate achievements & accept failures

24. Set goals in life

25. All of the above

26. Cleanliness

27. Maintaining personal hygiene for pleasing personality

28. smart

29. All of the above

30. She can wear it at night and wash her hair before leaving the house every day.

31. Information and Communication Technology

32. All of the above

33. . Inbox

34. Trash

35. Junk/Spam

36. Forward

37. Save as Draft

38. All of the above
39. User name & Domain name

40. Refresh Button

41. World Wide Web

42. Internet Service Provider

43. 1000 bits per second

44. Both a) and b)

45. Files

46. Society

47. Three Quarters

48. 5

49. Inexhaustible Resources

50. Renewable Resources

51. . Both a) and b)

52. . 2.5%

53. Natural Gas (CNG)

54. All of the above

55. Biodegradable Pollutants

56. Greenhouse Effect

57. Forest

58. 31 %

59. Chlorofluorocarbons

60. Chemical pollution due to fertilizers

61. Entrepreneurship

62. Both a) and b)

63. . Business

64. Small business

65. Both a) and b)

66. Positively

67. Characteristics of Entrepreneurship

68. New Ideas

69. Trial and error

70. Solution

71. 1989

72. Economic Growth

73. 2015

74. Space Research

75. Diesel